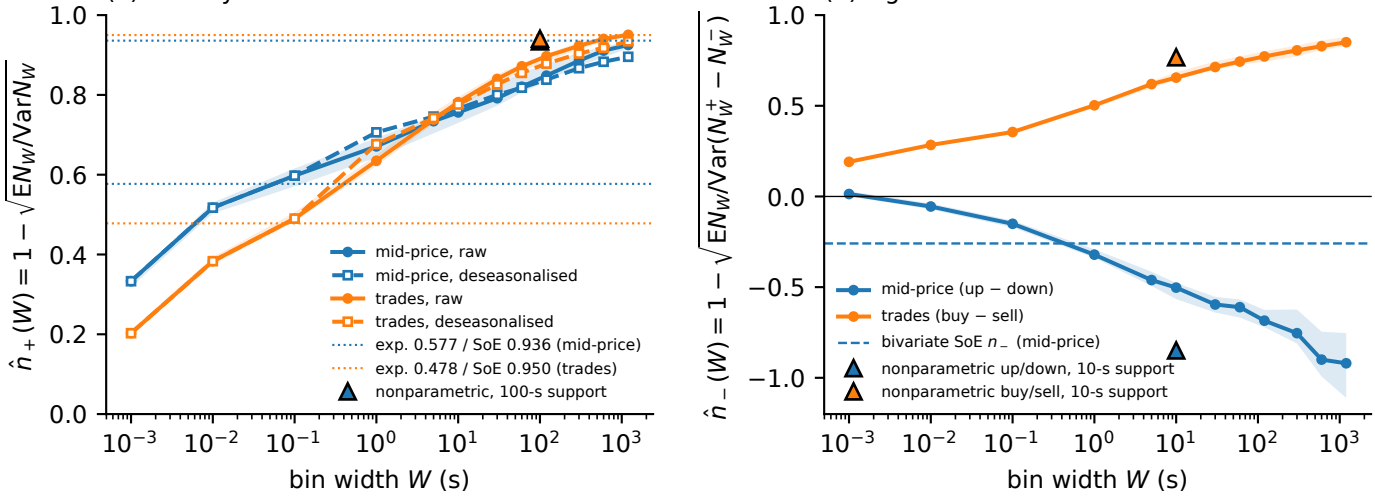
(a) activity mode
$\hat{n}_+(W) = 1 - \sqrt{EN_W/\mathrm{Var}N_W}$
1.0
0.8
0.6
0.4
0.2
0.0
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
bin width W (s)
mid-price, raw
mid-price, deseasonalised
trades, raw
trades, deseasonalised
exp. 0.577 / SoE 0.936 (mid-price)
exp. 0.478 / SoE 0.950 (trades)
nonparametric, 100-s support
(b) signed mode
$\hat{n}_-(W) = 1 - \sqrt{EN_W/\mathrm{Var}(N_W^+ - N_W^-)}$
1.0
0.5
0.0
−0.5
−1.0
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
bin width W (s)
mid-price (up − down)
trades (buy − sell)
bivariate SoE n_- (mid-price)
nonparametric up/down, 10-s support
nonparametric buy/sell, 10-s support